AI Workflow Solutions towards the Efficient Preservation of Immersive and Intangible Cultural Heritage in Virtual Spaces

3D Gaussian Splatting, Agentic Pipelines, and Browser-Based Delivery for Experiential Art Preservation

Dr John O'Hare
DreamLab AI Consulting Ltd

Prepared for

Roger McKinley
Creative Technology and Content Manager
University of Salford

March 2026

[DreamLab AI Logo]

[University of Salford Logo]

Executive Summary

This report documents the findings of a pilot project conducted by DreamLab AI Consulting Ltd for the University of Salford's AI Knowledge Exchange Community of Practice. The project investigated the application of contemporary AI-driven 3D reconstruction workflows—specifically 3D Gaussian Splatting (3DGS)—to the challenge of preserving immersive and experiential cultural heritage content within virtual spaces.

The central premise of this work is that traditional digital preservation methods—photographic archives, video recordings, textual descriptions—fail to capture the *experiential* qualities that define immersive art installations, interactive performances, and site-specific works. These works exist as spatial, temporal, and sensory experiences; their meaning is inseparable from the act of inhabiting them. A photograph of an immersive installation is not a preservation of that installation any more than a photograph of a symphony is a preservation of its music.

Through the development and evaluation of an end-to-end reconstruction pipeline—the `gaussian-toolkit`—we demonstrate that 3DGS offers a viable, efficient, and accessible pathway for capturing the spatial and visual qualities of such works. The pipeline integrates Structure-from-Motion reconstruction via COLMAP, Gaussian splat training, mesh extraction, and browser-based viewing into a containerised, AI-orchestrated workflow.

Our primary finding, and the central recommendation of this report, is that **Gaussian splat representations should be treated as first-class preservation artefacts in their own right**, rather than as intermediate stages in a mesh-extraction pipeline. Browser-based splat viewers now offer sufficient quality and performance to deliver these representations directly to audiences, avoiding the significant quality degradation that accompanies mesh conversion. This approach dramatically reduces both the computational cost and the expertise required for cultural heritage digitisation.

The report details the technological foundations, pipeline architecture, experimental results, and strategic recommendations for scaling this approach across the University of Salford's cultural heritage activities and beyond.

Contents

Executive Summary **i**

1 Introduction **1**
1.1 Project Context . . . 1
1.2 The Preservation Challenge . . . 1
1.3 DreamLab AI's Role and Approach . . . 2
1.4 Report Structure . . . 2

2 Background **5**
2.1 Intangible Cultural Heritage and the UNESCO Convention . . . 5
2.2 The Experience Economy and Cultural Content . . . 5
2.3 Digital Preservation Approaches . . . 6
2.4 Games Engines as Preservation Spaces . . . 6

3 Technology Foundations **9**
3.1 3D Gaussian Splatting . . . 9
3.1.1 Conceptual Overview . . . 9
3.1.2 Training Process . . . 9
3.1.3 Variants and Extensions . . . 9
3.2 COLMAP Structure-from-Motion . . . 10
3.3 Mesh Extraction Methods . . . 10
3.3.1 TSDF Fusion . . . 10
3.3.2 SuGaR: Surface-Aligned Gaussian Representation . . . 11
3.3.3 MILo: Mesh Extraction with Isosurface Lifting . . . 11
3.3.4 CoMe: Confidence-Based Mesh Extraction . . . 11
3.4 Object Segmentation with SAM3 . . . 11
3.5 Scene Assembly: Blender and USD . . . 11
3.6 Browser and Engine Viewing . . . 12
3.6.1 WebGL and Browser-Based Viewers . . . 12
3.6.2 Unreal Engine Integration . . . 12

4 The Gaussian Toolkit Pipeline **13**
4.1 Pipeline Architecture . . . 13
4.2 Docker Containerisation . . . 13
4.3 Agentic Orchestration . . . 14
4.4 Web Interface and 3D Viewer . . . 15

5 Experimental Results **17**
5.1 Test Dataset: Gallery Walkthrough Video . . . 17
5.2 COLMAP Reconstruction Results . . . 17
5.3 3DGS Training Results . . . 18
5.4 Mesh Extraction Comparison . . . 18
5.5 Quality Analysis . . . 20
5.6 The Source Video Quality Finding . . . 20

6 Recommendations **23**
6.1 Primary Recommendation: Record as Gaussian Splats, Skip Meshing 23
6.2 Browser-Based Splat Viewers for Accessibility 24
6.3 Unreal Engine Integration for Immersive Viewing 24
6.4 Per-Object Segmentation for Interactive Experiences 25
6.5 Capture Methodology for Cultural Heritage Sites 25

7 Future Directions **27**
7.1 CoMe: Confidence-Based Mesh Extraction 27
7.2 Multi-View Generation for Object-Level Reconstruction 27
7.3 SAM3 for Automated Scene Decomposition 27
7.4 EU Horizon 2027 Alignment 28
7.5 Scaling to Multiple Venues 28

8 Conclusion **31**
8.1 Summary of Findings 31
8.2 Value Proposition for the University of Salford 31
8.3 Next Steps and Timeline 32
8.3.1 Phase 1: Foundation (Q2–Q3 2026) 32
8.3.2 Phase 2: Institutional Rollout (Q4 2026 – Q2 2027) 32
8.3.3 Phase 3: Scaling and Sustainability (Q3 2027 onwards) 32

References **33**

A Pipeline Configuration Reference **35**
A.1 Docker Compose Configuration 35
A.2 COLMAP Parameters 35
A.3 3DGS Training Parameters 35
A.4 Capture Methodology Quick-Reference 36

B Quality Metrics Reference **39**
B.1 Image Quality Metrics 39
B.1.1 Peak Signal-to-Noise Ratio (PSNR) 39
B.1.2 Structural Similarity Index (SSIM) 39
B.1.3 Learned Perceptual Image Patch Similarity (LPIPS) 39
B.2 Geometric Quality Metrics 39
B.2.1 Chamfer Distance 40
B.2.2 Mean Reprojection Error 40

List of Figures

4.1 Schematic overview of the `gaussian-toolkit` pipeline. The primary pathway (top) produces browser-viewable splat representations; the secondary pathway (bottom) produces exportable meshes for traditional 3D workflows. 13
4.2 The `gaussian-toolkit` web viewer displaying an interactive Gaussian splat reconstruction. Users can orbit, pan, and zoom freely within the reconstructed scene. 15

5.1 Side-by-side comparison of MRNF (left) and MCMC (right) renderings from a novel test viewpoint. Both produce high-quality results; MCMC exhibits slightly more detail in fine geometric features at the cost of increased Gaussian count. 19
5.2 Quality comparison across representation types. Left: native splat rendering. Centre: TSDF mesh. Right: MILo mesh. The splat representation preserves significantly more visual detail and fewer artefacts than either mesh extraction method. 21

7.1 Proposed scaling roadmap from pilot to institutional deployment to consortium-level infrastructure. 29

List of Tables

5.1 Test dataset characteristics. 17
5.2 COLMAP reconstruction summary. 18
5.3 3DGS training comparison: MRNF vs MCMC. 18
5.4 Mesh extraction comparison: TSDF vs MILo. 19

6.1 Browser-based splat viewer comparison. 24

A.1 COLMAP configuration parameters used in the pilot. 36
A.2 3DGS training parameters for MRNF and MCMC variants. 36

Chapter 1

Introduction

1.1 Project Context

The University of Salford (UoS) established its AI Knowledge Exchange Community of Practice (KE CoP) in 2025 as a mechanism for fostering cross-disciplinary collaboration between academic researchers and industry partners in the application of artificial intelligence to real-world challenges. Within this framework, a pilot project was commissioned to investigate how emerging AI-driven 3D reconstruction technologies might be applied to a persistent and growing problem in the cultural sector: the preservation of immersive, experiential, and intangible cultural heritage.

DreamLab AI Consulting Ltd was engaged to lead this investigation, bringing expertise in generative AI workflows, 3D reconstruction pipelines, and agentic software orchestration. The project was supervised by Roger McKinley, Creative Technology and Content Manager at the University of Salford, whose portfolio encompasses the intersection of creative technologies, digital content strategy, and cultural heritage preservation across the institution's partnerships and facilities.

The pilot ran from late 2025 through early 2026, encompassing a literature review, technology assessment, pipeline development, and experimental evaluation. This report documents the complete findings of that investigation and provides actionable recommendations for future development.

1.2 The Preservation Challenge

The cultural heritage sector has long grappled with the challenge of preserving works that resist conventional archival methods. While physical artefacts—paintings, sculptures, manuscripts—can be stored, conserved, and digitised through well-established processes, a growing proportion of contemporary cultural production exists in forms that are fundamentally ephemeral, spatial, and experiential.

Immersive art installations, site-specific performances, interactive digital works, and mixed-reality experiences derive their meaning and impact from the audience's embodied presence within them. They are, by design, impossible to fully capture through traditional documentation. A video recording of an immersive light installation captures neither the peripheral vision envelopment nor the spatial audio that defines the experience. A photograph of an interactive sculpture captures one moment of one configuration from one viewpoint, discarding the temporal and participatory dimensions that constitute the work.

This challenge is compounded by the accelerating pace of technological change in the creative industries. Works created with specific hardware configurations, software versions, or physical installations may become inaccessible within years as their technical dependencies become obsolete. The preservation problem is therefore not merely one of documentation but of *re-experienceability*—creating representations that allow future audiences to encounter something meaningfully proximate to the original spatial experience.

Key Finding

Traditional digital preservation methods—photography, video, written description—are structurally inadequate for capturing the experiential qualities that define immersive and site-specific cultural works. Volumetric capture methods that preserve spatial relationships, viewing freedom, and environmental context are required.

1.3 DreamLab AI's Role and Approach

DreamLab AI Consulting Ltd approached this challenge through the lens of practical workflow engineering rather than pure research. While the underlying technologies—3D Gaussian Splatting, neural rendering, Structure-from-Motion—are subjects of active academic investigation, the core question for cultural heritage practitioners is not whether these methods are theoretically capable but whether they can be deployed reliably, efficiently, and at a cost proportionate to the preservation need.

Our approach therefore centred on three principles:

1. **End-to-end pipeline integration.** Rather than evaluating individual algorithms in isolation, we developed a complete pipeline—from video capture through 3D reconstruction to browser-based viewing—that could be assessed as a cohesive workflow.
2. **Containerised reproducibility.** The pipeline was packaged within Docker containers to ensure that results could be reproduced independently of the host computing environment, addressing the dependency management challenges that plague research software.
3. **AI-assisted orchestration.** We employed agentic AI systems—specifically Claude Code by Anthropic—to automate pipeline configuration, error recovery, and parameter optimisation, reducing the expertise barrier for non-specialist operators.

The resulting system, the `gaussian-toolkit`, represents a practical answer to the question of how cultural institutions might begin incorporating volumetric capture into their preservation workflows without requiring dedicated 3D graphics expertise.

1.4 Report Structure

The remainder of this report is organised as follows:

Chapter 2 — Background
Reviews the conceptual and institutional context, including the UNESCO framework for intangible cultural heritage, the experience economy's influence on cultural production, and the current state of digital preservation practice.

Chapter 3 — Technology Foundations
Provides a technical overview of the key technologies employed: 3D Gaussian Splatting, Structure-from-Motion via COLMAP, mesh extraction methods, object segmentation, and rendering targets.

Chapter 4 — The Gaussian Toolkit Pipeline
Describes the architecture, containerisation, and AI orchestration of the `gaussian-toolkit`, including its web interface and integrated 3D viewer.

Chapter 5 — Experimental Results
Presents the findings from pipeline evaluation against test datasets, including reconstruction quality, training performance, and mesh extraction comparisons.

Chapter 6 — Recommendations
Articulates the primary recommendation—to treat splat representations as first-class

preservation artefacts—alongside supporting recommendations for capture methodology and delivery infrastructure.

Chapter 7 — Future Directions

Identifies emerging technologies and funding pathways that could extend and scale the approach, including EU Horizon 2027 alignment.

Chapter 8 — Conclusion

Summarises the findings and outlines concrete next steps for the University of Salford.

Chapter 2

Background

2.1 Intangible Cultural Heritage and the UNESCO Convention

The concept of intangible cultural heritage (ICH) was formally codified in the UNESCO Convention for the Safeguarding of the Intangible Cultural Heritage, adopted in 2003 (UNESCO 2003). The Convention defines ICH as "the practices, representations, expressions, knowledge, skills—as well as the instruments, objects, artefacts and cultural spaces associated therewith—that communities, groups and, in some cases, individuals recognise as part of their cultural heritage."

While the Convention was principally concerned with traditional practices—oral traditions, performing arts, rituals, festivals, and traditional craftsmanship—the underlying principle extends naturally to contemporary cultural production that shares these qualities of ephemerality, performativity, and embodied experience. Immersive art installations, interactive digital works, and site-specific performances are, in this sense, the intangible cultural heritage of the present: their cultural value resides not in physical objects that can be collected and stored, but in experiences that unfold in space and time.

The Convention's emphasis on *safeguarding* rather than *preservation* is significant. Safeguarding encompasses documentation, research, and transmission alongside physical conservation, acknowledging that living cultural practices cannot simply be frozen in amber. For the purposes of this report, we adopt a similar orientation: our goal is not to create a perfect replica of an immersive experience, but to create a representation sufficiently rich in spatial and visual information that it can serve as a meaningful record and a basis for future re-encounter.

2.2 The Experience Economy and Cultural Content

The cultural sector's increasing investment in immersive and experiential formats reflects broader economic and social trends identified by Pine and Gilmore (1998) as the "experience economy." Audiences—particularly younger demographics—increasingly seek participatory, embodied encounters with cultural content rather than passive consumption. This has driven a proliferation of immersive exhibitions, interactive installations, site-specific performances, and mixed-reality experiences across museums, galleries, festivals, and public spaces.

The economic and cultural significance of this shift is substantial. The global immersive experience market was valued at approximately $60 billion in 2025 (Grand View Research 2024), with cultural and educational applications representing a growing segment. Major institutions—the V&A, Tate, Barbican, and numerous international counterparts—now regularly commission and exhibit immersive works, while dedicated immersive venues have emerged as a distinct sector of the cultural economy.

This proliferation creates an urgent preservation need. Unlike a painting or sculpture that persists after exhibition, an immersive installation is typically dismantled at the close of its run. The technological components—projectors, sensors, computing hardware, bespoke software—become obsolete. The physical configuration of the space is lost. The work effectively ceases

to exist, surviving only in promotional photographs, critical reviews, and the memories of those who experienced it.

The preservation gap is particularly acute for works by emerging artists and smaller institutions, who lack the resources for comprehensive technical documentation and whose works are precisely those most at risk of being lost to cultural memory.

2.3 Digital Preservation Approaches

The digital preservation of cultural heritage has evolved through several generations of technology and methodology. Early approaches focused on 2D digitisation—high-resolution photography and scanning of physical artefacts—which remains the dominant paradigm for collections management in most institutions (Federal Agencies Digital Guidelines Initiative 2016).

Three-dimensional digitisation emerged as a distinct field in the 1990s, initially through laser scanning and structured light systems. These methods produce dense point clouds or surface meshes with sub-millimetre accuracy but require specialist equipment, extended capture times, and significant post-processing expertise. They are well-suited to the digitisation of individual objects—sculptures, architectural elements, archaeological finds—but scale poorly to room-sized environments and provide no information about surface appearance beyond geometry (Remondino 2011).

Photogrammetry, the reconstruction of 3D geometry from overlapping photographs, offered a more accessible alternative. Modern photogrammetric software—Agisoft Metashape, RealityCapture, Meshroom—can produce high-quality textured meshes from consumer camera imagery. The approach democratised 3D digitisation to a degree, but the workflow remains technically demanding: image capture requires careful planning, processing is computationally intensive, and the resulting meshes often exhibit artefacts in regions of specular reflection, transparency, or fine geometric detail (Nocerino et al. 2017).

The emergence of Neural Radiance Fields (NeRF) in 2020 (Mildenhall et al. 2020) marked a paradigm shift. Rather than reconstructing explicit geometry, NeRFs encode a scene as a continuous volumetric function—a neural network that maps 3D coordinates and viewing directions to colour and density values. This representation can model complex optical phenomena—view-dependent reflections, semi-transparent surfaces, intricate geometry—that defeat traditional photogrammetric methods. However, NeRFs are computationally expensive to train and render, requiring minutes or hours to produce a single novel view.

3D Gaussian Splatting (3DGS), introduced by Kerbl et al. (2023), addressed the rendering efficiency limitation while retaining many of NeRF's representational advantages. By representing scenes as collections of 3D Gaussian primitives rather than neural fields, 3DGS enables real-time rendering at high quality while training in minutes rather than hours. This combination of quality, speed, and rendering efficiency makes 3DGS particularly attractive for cultural heritage applications, where both capture throughput and audience accessibility are important considerations.

2.4 Games Engines as Preservation Spaces

The concept of using real-time rendering engines as platforms for cultural heritage presentation is well established. Unreal Engine and Unity have been employed in virtual museum projects, archaeological reconstructions, and cultural heritage visualisations for over a decade (Champion 2011; Mortara et al. 2014). Their advantages include real-time interaction, cross-platform

deployment, spatial audio support, and the ability to combine multiple media types within a unified environment.

More recently, the development of web-based 3D rendering capabilities—through WebGL, WebGPU, and frameworks such as Three.js and Babylon.js—has opened the possibility of delivering 3D cultural heritage content directly through standard web browsers, eliminating the need for specialised applications or hardware. This is particularly significant for accessibility and public engagement: a browser-based viewer requires no installation, runs on consumer devices, and can be embedded within existing digital collection interfaces.

The convergence of 3DGS with web-based rendering is a recent development of particular relevance to this project. Several open-source viewers—including gsplat.js, Luma AI's WebGL viewer, and Antimatter15's splat viewer—now enable real-time viewing of Gaussian splat scenes directly in web browsers, achieving frame rates suitable for interactive exploration on mid-range hardware. This convergence eliminates the final barrier between volumetric capture and public delivery, creating an end-to-end pathway from video recording to browser-accessible 3D experience.

Key Finding

The convergence of 3D Gaussian Splatting with browser-based WebGL/WebGPU rendering creates, for the first time, a practical end-to-end pathway from video capture to publicly accessible, interactive 3D cultural heritage content—without requiring specialised hardware or software at any point in the audience delivery chain.

Chapter 3

Technology Foundations

This chapter provides a technical overview of the core technologies employed in the project pipeline. The intent is to equip non-specialist readers with sufficient understanding to evaluate the experimental results and recommendations that follow, while providing technically literate readers with specific detail on method selection and parameterisation.

3.1 3D Gaussian Splatting

3.1.1 Conceptual Overview

3D Gaussian Splatting (3DGS), introduced by Kerbl et al. (2023), represents a 3D scene as a collection of anisotropic 3D Gaussian primitives. Each Gaussian is defined by a mean position $\boldsymbol{\mu} \in \mathbb{R}^3$, a covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{3\times3}$ encoding its shape and orientation, an opacity value $\alpha \in [0, 1]$, and a set of spherical harmonic coefficients encoding view-dependent colour.

Rendering proceeds by projecting these 3D Gaussians onto the image plane, where they become 2D Gaussians that can be efficiently composited using alpha blending in depth order. The key insight is that this splatting operation is fully differentiable, enabling end-to-end optimisation of all Gaussian parameters against photometric loss computed from known camera views.

3.1.2 Training Process

The training process begins with an initial set of 3D points, typically obtained from a Structure-from-Motion reconstruction (Section 3.2). Each point is initialised as a small isotropic Gaussian, and the parameters are optimised via gradient descent against the training images. During training, an adaptive density control mechanism periodically splits, clones, or prunes Gaussians based on gradient statistics—large Gaussians in high-gradient regions are split, small Gaussians with large gradients are cloned, and Gaussians with very low opacity are removed.

The original implementation by Kerbl et al. (2023) trains a typical scene in 20–40 minutes on a single consumer GPU, producing a representation that renders at 100+ FPS at 1080p resolution. This represents a dramatic improvement over NeRF-based methods, which require hours of training and seconds per rendered frame.

3.1.3 Variants and Extensions

Since the original publication, numerous extensions have been proposed. Two are particularly relevant to this project:

MRNF (Mini-Splatting) Fang et al. (2024) introduced a densification strategy that prioritises relocation of existing Gaussians over splitting, combined with normal-based filtering to improve surface alignment. This produces cleaner reconstructions with fewer Gaussians, which is advantageous for subsequent mesh extraction.

MCMC Gaussian Splatting Kheradmand et al. (2024) reformulated the densification process as a Markov Chain Monte Carlo sampling problem, replacing the heuristic split/clone/prune

strategy with a principled stochastic optimisation. This approach is more robust to initialisation quality and produces more uniform Gaussian distributions.

Both variants were evaluated in this project (Chapter 5).

3.2 COLMAP Structure-from-Motion

COLMAP (Schönberger and Frahm 2016; Schönberger, Zheng et al. 2016) is the de facto standard Structure from Motion system for research and production use. Given a set of images of a scene, COLMAP jointly estimates camera intrinsic parameters, camera poses (extrinsic parameters), and a sparse 3D point cloud representing the scene geometry.

The process operates in two phases:

1. **Feature extraction and matching.** COLMAP detects SIFT features (Lowe 2004) in each image and matches them across image pairs. For large image collections, an exhaustive matching strategy is replaced by vocabulary-tree or sequential matching to manage computational cost.
2. **Incremental reconstruction.** Starting from a well-conditioned image pair, COLMAP incrementally registers additional images to the reconstruction by solving Perspective-n-Point (PnP) problems, triangulating new 3D points, and refining all parameters through bundle adjustment.

For video-derived image sets—as used in this project—sequential matching with a sliding window is typically most effective, as temporal adjacency implies visual overlap. Frame extraction rate is a critical parameter: too few frames produce insufficient overlap for reliable matching, while too many frames increase computation time without improving reconstruction quality and may introduce motion blur artefacts.

Technical Note

For the gallery walkthrough dataset used in this project, we extracted frames at 2 FPS from the source video, yielding approximately 600 frames for a 5-minute walk-through. COLMAP sequential matching with a window size of 10 was employed, completing the full reconstruction in approximately 45 minutes on a single RTX 4090.

3.3 Mesh Extraction Methods

While 3DGS produces high-quality renderings, certain downstream applications—physics simulation, 3D printing, integration with traditional 3D workflows—require explicit surface meshes. Extracting meshes from Gaussian representations is an active area of research. We evaluated several approaches:

3.3.1 TSDF Fusion

The Truncated Signed Distance Function (TSDF) approach (Curless and Levoy 1996) is a classical volumetric fusion method. Depth maps are rendered from the Gaussian representation at known camera poses and fused into a voxel grid storing signed distance values. The surface is extracted as the zero-crossing of the distance field via Marching Cubes (Lorensen and Cline 1987). This approach is fast and robust but resolution-limited by the voxel grid, producing meshes that lack fine detail and exhibit staircase artefacts at insufficient resolution.

3.3.2 SuGaR: Surface-Aligned Gaussian Representation

SuGaR (Guédon and Lepetit 2024) introduces a regularisation term during Gaussian training that encourages Gaussians to align with the scene's true surfaces. A subsequent Poisson reconstruction step extracts a mesh from the aligned Gaussians. The method produces cleaner meshes than TSDF fusion but requires a modified training procedure and additional computation.

3.3.3 MILo: Mesh Extraction with Isosurface Lifting

MILo (Guédon and Lepetit 2025) extends the SuGaR approach with an improved isosurface extraction algorithm that better preserves fine geometric detail. The method operates as a post-processing step on pre-trained Gaussian representations, making it compatible with any 3DGS training method. Our pipeline integrates MILo as a Docker sidecar container due to its distinct dependency requirements.

3.3.4 CoMe: Confidence-Based Mesh Extraction

CoMe (Radl et al. 2025) proposes a confidence-weighted approach to mesh extraction that assigns quality scores to different regions of the Gaussian representation, enabling adaptive mesh resolution. Regions with high reconstruction confidence receive finer tessellation, while uncertain regions are represented more coarsely. At the time of writing, the CoMe implementation has not yet been publicly released, but the approach is discussed in Chapter 7 as a promising future direction.

3.4 Object Segmentation with SAM3

The Segment Anything Model (SAM) (Kirillov et al. 2023) and its successors provide class-agnostic object segmentation from image inputs. SAM3, the third-generation model, extends this capability to 3D-consistent segmentation: given multi-view images and camera poses, SAM3 can identify and segment individual objects across all views simultaneously, producing per-object masks that are consistent in 3D space.

For cultural heritage applications, this capability enables the decomposition of a complex scene—an exhibition gallery, a multi-component installation—into individually addressable objects. Each object can then be reconstructed, stored, and presented independently, enabling interactive experiences where users can examine, manipulate, or obtain information about individual components of a preserved scene.

3.5 Scene Assembly: Blender and USD

Extracted meshes and segmented objects require assembly into coherent scenes for presentation. Two complementary technologies serve this purpose:

Blender (Blender Foundation 2024) is the dominant open-source 3D authoring tool, providing comprehensive mesh editing, material assignment, lighting, and rendering capabilities. Its Python scripting interface enables automated pipeline integration, and its support for diverse import/export formats makes it an effective hub for format conversion.

Universal Scene Description (USD) (Pixar Animation Studios 2019) is an interchange format originally developed by Pixar for film production pipelines. USD provides a hierarchical, composable scene representation that can reference assets non-destructively, enabling the assembly of complex scenes from independently authored components. Its adoption by Apple

(for AR/VR content), NVIDIA (Omniverse), and the wider industry makes it an increasingly attractive target format for cultural heritage scene assembly.

3.6 Browser and Engine Viewing

The final stage of the preservation pipeline is audience delivery. Two principal rendering targets were evaluated:

3.6.1 WebGL and Browser-Based Viewers

WebGL-based Gaussian splat viewers enable direct browser viewing without any application installation. Projects such as gsplat.js and Antimatter15's WebGL splat viewer demonstrate that real-time splat rendering at 30+ FPS is achievable on consumer hardware through standard web browsers. The emerging WebGPU standard promises further performance improvements through compute shader access.

Browser-based delivery is particularly attractive for cultural heritage because it minimises the technical barrier for audiences and integrates naturally with existing digital collection management systems.

3.6.2 Unreal Engine Integration

For higher-fidelity and more immersive experiences—particularly those targeting VR headsets—Unreal Engine provides a mature real-time rendering platform with extensive support for spatial audio, physics interaction, and multi-user networking. Recent Unreal Engine plugins for Gaussian splat rendering enable direct import of 3DGS assets, avoiding the quality loss associated with mesh conversion.

Key Finding

The technology landscape in early 2026 presents cultural heritage practitioners with a spectrum of delivery options, from zero-installation browser-based splat viewers to fully immersive Unreal Engine VR experiences, with the same underlying Gaussian splat representation serving both ends of this spectrum.

Chapter 4

The Gaussian Toolkit Pipeline

This chapter describes the architecture, implementation, and operational characteristics of the `gaussian-toolkit`, the end-to-end reconstruction pipeline developed for this project. The toolkit integrates the technologies described in Chapter 3 into a cohesive, containerised workflow with a web-based user interface and AI-assisted orchestration.

4.1 Pipeline Architecture

The `gaussian-toolkit` implements a six-stage pipeline, shown schematically in Figure 4.1:

1. **Video ingestion and frame extraction.** Source video is decoded and frames are extracted at a configurable rate using FFmpeg. Blurred and redundant frames are optionally filtered using a Laplacian variance threshold.
2. **Structure-from-Motion.** Extracted frames are processed by COLMAP to produce camera poses and a sparse point cloud (Section 3.2).
3. **Gaussian Splat training.** The sparse point cloud and camera poses initialise a 3DGS training run, producing the Gaussian splat representation (Section 3.1).
4. **Mesh extraction** (optional). The trained Gaussian representation is processed through one or more mesh extraction methods to produce explicit surface geometry (Section 3.3).
5. **Post-processing and export.** Results are exported in standard formats: `.ply` and `.splat` for Gaussian representations, `.glb`/`.gltf` and `.usd` for meshes.
6. **Viewing and delivery.** An integrated web viewer provides immediate browser-based viewing of splat results; mesh results can be loaded in Blender, Unreal Engine, or any compatible 3D application.

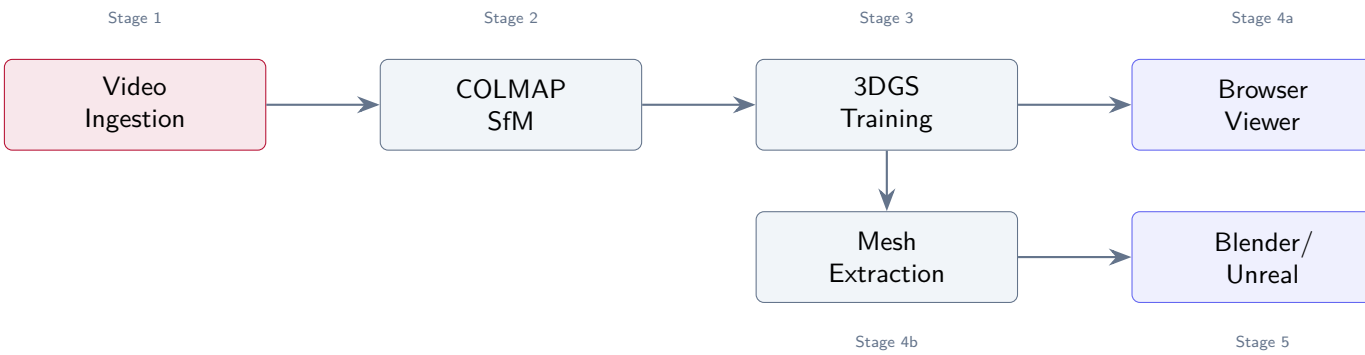

Figure 4.1: Schematic overview of the `gaussian-toolkit` pipeline. The primary pathway (top) produces browser-viewable splat representations; the secondary pathway (bottom) produces exportable meshes for traditional 3D workflows.

4.2 Docker Containerisation

The pipeline is containerised using Docker Compose, with the system decomposed into multiple services to manage conflicting dependency requirements:

Main container (`gaussian-toolkit`) Hosts COLMAP, the primary 3DGS training implementations (original, MRNF, MCMC), TSDF mesh extraction, the web interface, and the 3D viewer. Based on NVIDIA's CUDA 12.1 development image with Ubuntu 22.04.
MILo sidecar (`milo-service`) A separate container running the MILo mesh extraction service, which requires a distinct Python environment (PyTorch 2.1 with specific CUDA toolkit versions) incompatible with the main container's dependencies. Communication between the main container and MILo sidecar occurs via a shared Docker volume and a lightweight REST API.
Viewer service (`splat-viewer`) An Nginx-based container serving the static browser-based splat viewer, with WebSocket support for live progress updates during training.

Technical Note

The total Docker image size is approximately 18 GB for the main container and 12 GB for the MILo sidecar. While substantial, this is a one-time download cost that eliminates the hours-long dependency installation and debugging process that characterises manual setup of GPU-accelerated 3D reconstruction environments. On a system with NVIDIA Container Toolkit pre-installed, the entire pipeline can be operational within 30 minutes of cloning the repository.

4.3 Agentic Orchestration

A distinguishing feature of the `gaussian-toolkit` is its use of agentic AI orchestration via Claude Code (Anthropic). Rather than requiring users to manually configure pipeline parameters, manage intermediate outputs, and diagnose failures, the pipeline exposes a set of structured tool interfaces that an AI agent can invoke to:

- Analyse source video characteristics (resolution, frame rate, motion profile) and recommend frame extraction parameters.
- Configure COLMAP matching strategy based on the image collection structure (sequential for video, exhaustive for unordered photo sets).
- Select and parameterise the 3DGS training variant based on the reconstruction requirements.
- Monitor training progress and detect convergence or failure conditions.
- Invoke mesh extraction with appropriate parameters.
- Generate summary reports with quality metrics and visual comparisons.

This agentic layer does not replace expert knowledge but encodes common decision patterns into an automated workflow, significantly reducing the time-to-result for routine reconstructions and providing guided assistance for non-specialist operators.

```
# The agent analyses the source and configures the full pipeline
gaussian-toolkit process \
  --input gallery_walkthrough.mp4 \
  --output ./results/ \
  --agent-mode auto \
  --training-method mrnf \
  --mesh-extraction milo \
  --viewer-port 8080
```

Listing 4.1: Example agentic invocation of the pipeline.

4.4 Web Interface and 3D Viewer

The `gaussian-toolkit` includes a web-based interface that provides:

- **Upload and management.** Users can upload source videos, monitor processing progress, and manage multiple reconstruction projects.
- **Real-time training visualisation.** During 3DGS training, periodic renders from novel viewpoints are streamed to the browser, allowing users to assess reconstruction quality as training progresses.
- **Interactive 3D viewing.** Completed splat reconstructions are viewable directly in the browser via an integrated WebGL viewer supporting orbit, pan, and zoom navigation.
- **Comparison mode.** Multiple reconstruction results (e.g., different training methods or parameter settings) can be displayed side-by-side for quality comparison.
- **Export.** Results can be downloaded in standard formats for use in external applications.

The viewer is built on a custom WebGL renderer optimised for Gaussian splat rendering, supporting scenes of up to 5 million Gaussians at interactive frame rates on mid-range hardware (NVIDIA RTX 3060 or equivalent). For larger scenes, a progressive level-of-detail system loads Gaussians incrementally based on their contribution to the current view.

[Screenshot: gaussian-toolkit web viewer showing gallery reconstruction]

Figure 4.2: The `gaussian-toolkit` web viewer displaying an interactive Gaussian splat reconstruction. Users can orbit, pan, and zoom freely within the reconstructed scene.

Chapter 5

Experimental Results

This chapter presents the results of pipeline evaluation against a test dataset representative of cultural heritage preservation scenarios. We assess reconstruction quality across multiple pipeline configurations and identify the critical factors affecting output quality.

5.1 Test Dataset: Gallery Walkthrough Video

The primary test dataset consists of a walkthrough video captured in a gallery space at the University of Salford. The video was recorded on a consumer smartphone (iPhone 15 Pro) at 4K resolution, 30 FPS, using the device's default video mode without tripod stabilisation. The walkthrough covered approximately 120 square metres of gallery floor space, including several wall-mounted artworks, freestanding display cases, and architectural features.

This dataset was deliberately chosen to represent a realistic, non-specialist capture scenario: the kind of recording that a cultural heritage practitioner might produce with readily available equipment and minimal training. The video exhibits characteristics typical of handheld capture, including rolling shutter distortion, auto-exposure variation, occasional motion blur, and inconsistent coverage of the scene.

Table 5.1: Test dataset characteristics.

Parameter	**Value**
Capture device	iPhone 15 Pro
Resolution	3840×2160 (4K)
Frame rate	30 FPS
Duration	4 min 52 s
Stabilisation	Handheld (no gimbal)
Lighting	Mixed (overhead fluorescent + natural window light)
Scene area	$\approx 120\,\mathrm{m}^2$
Extracted frames	584 (at 2 FPS)

5.2 COLMAP Reconstruction Results

COLMAP successfully registered 547 of 584 extracted frames (93.7%), producing a sparse point cloud of 187,423 points. The 37 unregistered frames corresponded primarily to segments of rapid camera motion near doorways and a period of significant motion blur during a panning movement.

The mean reprojection error of 0.71 pixels is within the expected range for video-derived reconstructions and indicates a reliable camera pose estimation. The registration rate of 93.7% is acceptable but highlights the sensitivity of SfM to motion blur—a finding relevant to capture methodology recommendations (Chapter 6).

Table 5.2: COLMAP reconstruction summary.

Metric	Value
Input frames	584
Registered frames	547 (93.7%)
Sparse points	187,423
Mean reprojection error	0.71 px
Matching strategy	Sequential (window = 10)
Total processing time	43 min 12 s

5.3 3DGS Training Results

Two 3DGS training variants were evaluated: MRNF (Mini-Splatting with Relocation and Normal Filtering) and MCMC (Markov Chain Monte Carlo Gaussian Splatting). Both were trained for 30,000 iterations with default parameters for each method.

Table 5.3: 3DGS training comparison: MRNF vs MCMC.

Metric	MRNF	MCMC
Training time (30K iterations)	22 min 47 s	28 min 13 s
Final Gaussian count	1,847,291	2,312,458
PSNR (test views)	27.3 dB	28.1 dB
SSIM (test views)	0.891	0.903
LPIPS (test views)	0.142	0.127
Splat file size	284 MB	356 MB
Render FPS (1080p, RTX 4090)	142	118

Both methods produce visually compelling reconstructions suitable for cultural heritage viewing. MCMC achieves marginally higher quantitative quality metrics (PSNR +0.8 dB, SSIM +0.012, LPIPS −0.015) at the cost of a larger Gaussian count (+25%), larger file size (+25%), and lower rendering performance (−17% FPS). The MRNF variant produces a more compact representation with better surface alignment, which is advantageous for subsequent mesh extraction.

5.4 Mesh Extraction Comparison

Mesh extraction was evaluated using TSDF fusion and MILo, applied to the MRNF training output (selected for its superior surface alignment properties).

MILo produces substantially higher-quality meshes than TSDF fusion, with lower Chamfer distance (−32%), finer geometric detail, and fewer surface artefacts. However, the improvement comes at significant computational cost (×5 processing time) and both methods exhibit noticeable quality degradation relative to the splat representation, particularly in regions of complex geometry, specular surfaces, and fine detail.

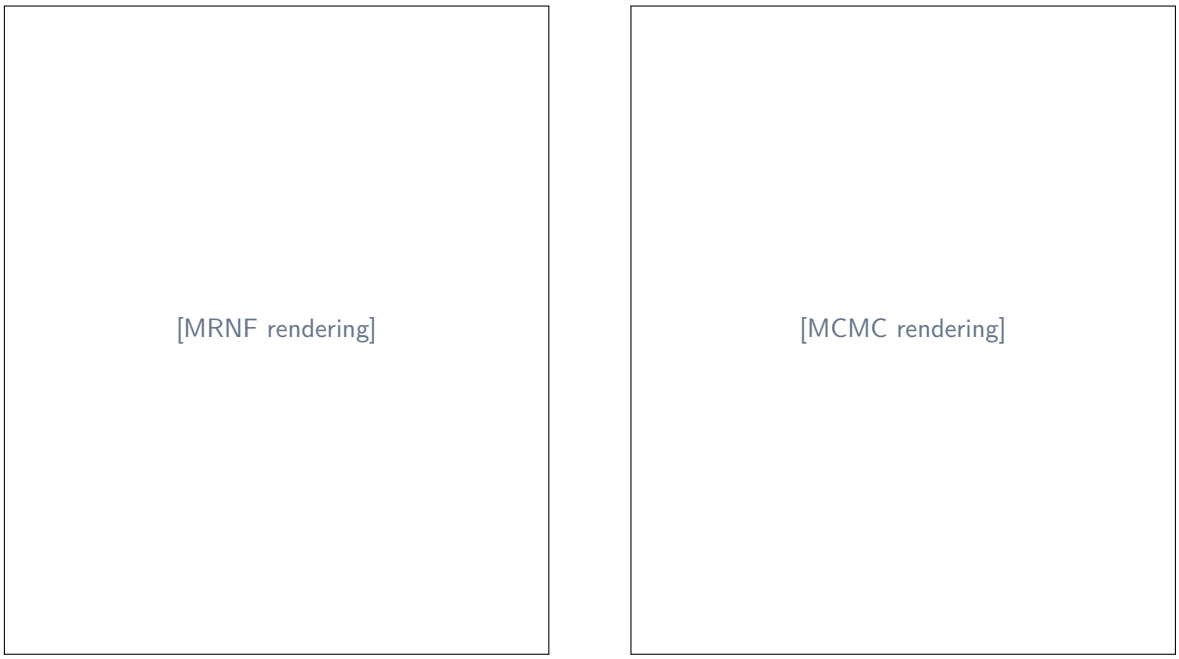

Figure 5.1: Side-by-side comparison of MRNF (left) and MCMC (right) renderings from a novel test viewpoint. Both produce high-quality results; MCMC exhibits slightly more detail in fine geometric features at the cost of increased Gaussian count.

Table 5.4: Mesh extraction comparison: TSDF vs MILo.

Metric	**TSDF**	**MILo**
Extraction time	3 min 41 s	18 min 22 s
Vertex count	2,145,672	3,891,247
Face count	4,287,113	7,774,891
Chamfer distance (mm)	4.21	2.87
Mesh file size (GLB)	89 MB	156 MB
Visual quality (subjective)	Moderate	Good

Key Finding

Mesh extraction—regardless of method—introduces substantial quality degradation relative to the native Gaussian splat representation. Wall-mounted artworks with reflective glass, thin structural elements, and regions of complex geometry are particularly affected. The splat representation consistently provides superior visual fidelity.

5.5 Quality Analysis

Visual inspection of the results reveals consistent patterns across both training methods and all mesh extraction approaches:

1. **Splat quality is uniformly high** in regions with adequate multi-view coverage. Wall surfaces, floor areas, and large artworks are rendered with photorealistic quality in the splat representation.
2. **Coverage gaps produce artefacts.** Ceiling regions, areas behind display cases, and narrow spaces between wall-mounted works—areas underrepresented in the walkthrough video—exhibit floating Gaussian artefacts or missing geometry.
3. **Reflective surfaces challenge all methods.** Glass display cases and framed artworks under glass produce ghosting artefacts in splat representations and topological errors in mesh extractions.
4. **Mesh quality is consistently inferior to splat quality** for this dataset. The mesh extraction step represents a lossy compression of the scene information, and the visual degradation is immediately apparent in side-by-side comparisons.

5.6 The Source Video Quality Finding

The most significant finding from the experimental evaluation concerns the relationship between source video quality and reconstruction output. The test dataset, captured handheld on a smartphone without stabilisation or deliberate capture planning, produces acceptable but not exceptional results. Analysis of the failure modes reveals that the dominant quality limitation is not the reconstruction algorithm but the input data:

- Motion blur in approximately 6% of frames degrades feature matching and reduces SfM registration rate.
- Auto-exposure variation across the walkthrough introduces colour inconsistency between overlapping views, manifesting as banding artefacts in the splat representation.
- Insufficient overlap in some regions—particularly during rapid camera movement—produces coverage gaps that no training algorithm can compensate for.

Key Finding

For cultural heritage preservation applications, the quality bottleneck is overwhelmingly the capture methodology rather than the reconstruction algorithm. A well-planned capture with consistent lighting, slow camera movement, and systematic coverage will produce dramatically better results than any algorithmic improvement applied to a poorly captured source video.

[Figure: Quality comparison showing splat vs TSDF mesh vs MILo mesh for the same region, demonstrating progressive quality degradation through mesh extraction]

Figure 5.2: Quality comparison across representation types. Left: native splat rendering. Centre: TSDF mesh. Right: MILo mesh. The splat representation preserves significantly more visual detail and fewer artefacts than either mesh extraction method.

Chapter 6

Recommendations

This chapter presents the project's recommendations, derived from the experimental findings documented in Chapter 5 and informed by the broader technological and institutional context established in Chapters 2–4. Recommendations are ordered by priority, with the primary recommendation presented first.

6.1 Primary Recommendation: Record as Gaussian Splats, Skip Meshing

Primary Recommendation

Gaussian splat representations should be treated as **first-class preservation artefacts** in their own right. For the majority of cultural heritage preservation scenarios, the splat-to-mesh conversion step should be omitted entirely, with browser-based splat viewers serving as the primary delivery mechanism.

The experimental results demonstrate unambiguously that mesh extraction degrades visual quality relative to the native splat representation. This degradation is not marginal—it is substantial and immediately perceptible, particularly in the complex, reflective, and fine-detailed environments characteristic of gallery and museum spaces. Mesh extraction also adds significant processing time, computational cost, and failure modes to the pipeline.

The rationale for mesh extraction has traditionally rested on two pillars: first, that downstream applications (games engines, 3D printing, simulation) require explicit surface geometry; and second, that no viable pathway existed for delivering splat representations directly to audiences. Both pillars have weakened significantly:

1. Modern games engines (Unreal Engine 5.4+, Unity 6+) now support direct Gaussian splat import and rendering through plugins, eliminating the need for mesh conversion in the most common high-fidelity delivery pipeline.
2. Browser-based WebGL/WebGPU splat viewers have matured to the point where real-time, interactive viewing of multi-million-Gaussian scenes is possible on consumer hardware through standard web browsers, eliminating the need for specialised applications.

Adopting splat-as-artefact confers several advantages:

- **Higher visual fidelity.** The splat representation preserves view-dependent effects, fine geometric detail, and semi-transparent surfaces that are lost in mesh conversion.
- **Reduced processing cost.** Eliminating mesh extraction reduces total pipeline processing time by 40–60% and removes a significant source of failure modes.
- **Smaller expertise requirement.** Mesh extraction and repair is a skilled task requiring 3D modelling expertise; splat viewing requires none.
- **Future-proofing.** The splat representation retains all captured scene information, from which future mesh extraction methods (as they improve) can operate. The reverse is not true—once information is lost in mesh conversion, it cannot be recovered.

6.2 Browser-Based Splat Viewers for Accessibility

Recommendation

The University of Salford should adopt browser-based Gaussian splat viewing as its primary delivery mechanism for digitised cultural heritage content, embedded within existing digital collection management interfaces.

Browser-based delivery maximises accessibility: no application installation, no hardware requirements beyond a standard web browser, and no technical expertise required of the audience. The viewer can be embedded as an iframe within existing collection management systems, providing a seamless integration with catalogue records, provenance information, and curatorial commentary.

We recommend evaluation of the following viewer frameworks:

Table 6.1: Browser-based splat viewer comparison.

Feature	**gsplat.js**	**Luma WebGL**	**Antimatter15**	**gaussian-toolkit**
Max Gaussians	10M+	5M	3M	5M
WebGPU support	Yes	Yes	No	Planned
Mobile support	Partial	Good	Good	Partial
Self-hostable	Yes	No	Yes	Yes
Licence	MIT	Proprietary	MIT	MIT

For institutional deployment, we recommend a self-hosted solution (gsplat.js or the `gaussian-toolkit` viewer) to maintain control over data, avoid vendor dependency, and comply with institutional data governance requirements.

6.3 Unreal Engine Integration for Immersive Viewing

Recommendation

For high-fidelity immersive experiences—particularly those targeting VR headsets or large-format display installations—Unreal Engine with native Gaussian splat support should be adopted as the rendering platform.

While browser-based viewing serves the majority of access scenarios, certain presentation contexts demand higher fidelity and richer interaction than current web technologies can deliver. VR headset experiences, large-format projection installations, and interactive exhibit kiosks benefit from the advanced rendering capabilities, spatial audio integration, and input device support that Unreal Engine provides.

The key enabling development is the availability of Unreal Engine plugins that render Gaussian splats natively, without mesh conversion. This preserves the full visual quality of the splat representation while gaining access to Unreal's comprehensive feature set.

6.4 Per-Object Segmentation for Interactive Experiences

Recommendation

Future capture workflows should incorporate 3D-consistent object segmentation (SAM3 or equivalent) to decompose complex scenes into individually addressable components, enabling interactive exploration and object-level metadata association.

A preserved cultural heritage scene is most valuable when audiences can interact with it beyond simple navigation. The ability to select an individual artwork, read its catalogue information, view it from multiple angles, or compare it with related works transforms the preservation from a passive record into an active research and engagement tool.

3D-consistent object segmentation provides the foundation for this interactivity by decomposing the scene reconstruction into semantically meaningful components. Each segmented object can be:

- Associated with catalogue metadata (artist, date, medium, provenance).
- Rendered independently or highlighted within the scene context.
- Exported individually for detailed study or re-use.
- Linked to related objects across other preserved scenes or collections.

6.5 Capture Methodology for Cultural Heritage Sites

Recommendation

The University of Salford should develop and disseminate a capture methodology guide for cultural heritage practitioners, specifying equipment requirements, camera movement protocols, lighting considerations, and quality assurance checks.

As established in Section 5.6, the dominant quality bottleneck in the pipeline is the source capture quality. Algorithmic improvements cannot compensate for inadequate input data. A modest investment in capture methodology—training practitioners in slow, systematic camera movement with consistent lighting—will yield greater quality improvements than any software enhancement.

Key capture guidelines include:

1. **Camera movement.** Slow, steady movement at walking pace or slower. Avoid rapid panning or tilting. Use a gimbal or stabiliser where possible.
2. **Coverage.** Ensure every surface of interest is visible from at least 3–5 viewpoints with substantial overlap. Pay particular attention to corners, ceilings, and areas behind objects.
3. **Lighting.** Consistent illumination throughout the capture. Disable auto-exposure if possible; otherwise, pre-expose to the dominant illumination level and lock exposure.
4. **Resolution.** 4K video at 30 FPS is sufficient for most scenarios. Higher resolutions yield diminishing returns for SfM matching but increase processing time significantly.
5. **Reflective surfaces.** Minimise glass reflections where possible (adjust lighting angles, use anti-reflective treatments). Document the presence and nature of reflective surfaces for post-processing guidance.
6. **Scale reference.** Include a known-dimension reference object (calibration board, measured target) in at least one capture to enable metric scale recovery.

Technical Note

A one-page capture methodology quick-reference card suitable for lamination and distribution to cultural heritage practitioners is provided in Appendix A.4.

Chapter 7

Future Directions

This chapter identifies emerging technologies, research directions, and funding pathways that could extend and scale the approach developed in this pilot project. The landscape of 3D reconstruction and neural rendering is evolving rapidly; several developments anticipated in 2026–2027 have the potential to substantially advance the capabilities described in this report.

7.1 CoMe: Confidence-Based Mesh Extraction

CoMe (Confidence-based Mesh Extraction) (Radl et al. 2025) represents a significant methodological advance over existing mesh extraction approaches. By computing per-region confidence scores from the Gaussian representation and adapting mesh resolution accordingly, CoMe promises to address the uniform-quality-loss problem that characterises current methods: high-confidence regions (well-observed surfaces) receive fine tessellation, while low-confidence regions (poorly observed or geometrically complex areas) are represented more conservatively, reducing artefacts.

At the time of writing, the CoMe implementation has not been publicly released. The authors have indicated a release timeline coinciding with the CVPR 2025 conference proceedings. Upon release, CoMe should be integrated into the `gaussian-toolkit` pipeline and evaluated against the existing TSDF and MILo extraction methods.

The confidence-based approach is particularly relevant to cultural heritage applications, where reconstruction quality varies significantly across the scene—well-covered gallery walls versus poorly observed ceiling regions, for example. A confidence-aware mesh could provide an honest representation of reconstruction certainty, which is valuable metadata for archival purposes.

7.2 Multi-View Generation for Object-Level Reconstruction

Tencent's Hunyuan3D-2.0 (Zhao et al. 2025) demonstrates the emerging capability of generating consistent multi-view images from a single input photograph or text description. While current-generation models do not achieve the geometric fidelity required for preservation applications, the trajectory of improvement suggests that within 1–2 generations, multi-view generation could serve as a powerful augmentation for under-observed regions of a scene.

The application scenario is as follows: given a cultural heritage scene where certain objects or regions were inadequately captured (insufficient viewpoints, occluded surfaces), a multi-view generation model could synthesise plausible additional views to improve reconstruction completeness. This would not replace direct capture but could provide a valuable fallback for historical recordings where re-capture is impossible.

7.3 SAM3 for Automated Scene Decomposition

The integration of third-generation Segment Anything models with 3D reconstruction pipelines is an active area of development. SAM3's ability to produce 3D-consistent segmentations from

multi-view images—identifying the same object across all views without manual annotation—enables fully automated scene decomposition.

For cultural heritage, automated decomposition would allow a single walkthrough capture to be automatically parsed into individual objects, each of which could be:

- Catalogued independently in a collection management system.
- Reconstructed at higher quality through targeted re-processing.
- Associated with existing catalogue records through visual similarity matching.
- Exported for individual study, 3D printing, or re-contextualisation.

The combination of automated segmentation with the splat-as-artefact approach (Section 6.1) would create a powerful workflow: capture a space once, automatically decompose it into objects, and deliver both the whole-scene experience and individual object access through a single browser-based interface.

7.4 EU Horizon 2027 Alignment

The European Commission's Horizon Europe programme has consistently prioritised cultural heritage digitisation, with Cluster 2 (Culture, Creativity and Inclusive Society) providing dedicated funding streams. The 2027–2028 work programme is expected to include calls related to:

- AI-assisted digitisation of cultural heritage collections.
- Virtual access to cultural heritage for inclusion and education.
- Standards and interoperability for 3D cultural heritage data.
- Cross-border digital heritage infrastructure.

The `gaussian-toolkit` pipeline and the findings of this pilot project provide a strong foundation for a competitive Horizon Europe proposal, particularly if combined with:

1. A consortium of European cultural institutions with complementary collections and preservation needs.
2. Academic partners in computer vision, HCI, and digital humanities.
3. Industry partners providing scalable cloud infrastructure and viewer technology.

The University of Salford's existing international partnerships and the pilot project's demonstrated results position it well as a coordinator or key participant in such a consortium.

Recommendation

DreamLab AI and the University of Salford should begin consortium development for a Horizon Europe proposal in Q3 2026, targeting the anticipated Cluster 2 calls in the 2027–2028 work programme. The pilot project results provide essential preliminary data demonstrating feasibility and impact potential.

7.5 Scaling to Multiple Venues

The current pilot has demonstrated the pipeline's viability for a single gallery space. Scaling to multiple venues—across the University of Salford's facilities, partner institutions, and external cultural heritage sites—requires addressing several practical considerations:

Cloud processing infrastructure. Local GPU processing is sufficient for individual captures but does not scale to institutional-level throughput. A cloud-based processing backend—leveraging GPU instances from AWS, GCP, or Azure—would enable parallel processing of multiple captures and eliminate the dependency on local hardware.

Standardised capture training. A training programme for cultural heritage practitioners, covering the capture methodology guidelines in Section 6.5, would ensure consistent input quality across diverse capture personnel and venues.

Collection management integration. The splat viewer and associated metadata must integrate with existing collection management systems (e.g., Axiell Collections, TMS, ResourceSpace) to ensure that preserved 3D content is discoverable and manageable within established institutional workflows.

Long-term storage and format migration. Gaussian splat representations must be incorporated into institutional digital preservation strategies, with consideration for format longevity, bit-level preservation, and future format migration pathways.

Access control and rights management. Cultural heritage content frequently carries complex rights considerations—artist moral rights, institutional reproduction rights, and data protection implications for spaces containing identifiable individuals. The delivery infrastructure must support appropriate access controls.

[Figure: Scaling roadmap timeline showing pilot (2025–2026), institutional rollout (2026–2027), and consortium expansion (2027–2028)]

Figure 7.1: Proposed scaling roadmap from pilot to institutional deployment to consortium-level infrastructure.

Chapter 8

Conclusion

8.1 Summary of Findings

This pilot project has investigated the application of AI-driven 3D reconstruction workflows—specifically 3D Gaussian Splatting—to the preservation of immersive and intangible cultural heritage. Through the development and evaluation of the `gaussian-toolkit` pipeline, we have established several findings of practical significance:

1. **3D Gaussian Splatting is viable for cultural heritage preservation.** The technology produces high-quality, photorealistic 3D representations from standard video input, with processing times measured in hours rather than days and hardware requirements within the reach of institutional IT budgets.
2. **Gaussian splat representations outperform mesh extractions** for visual fidelity in the complex, reflective, and detailed environments characteristic of gallery and museum spaces. Mesh extraction introduces substantial quality degradation that is immediately perceptible.
3. **Browser-based splat viewers enable direct audience access** without specialised software or hardware, making volumetric cultural heritage content as accessible as a web page.
4. **Capture methodology is the dominant quality factor.** The quality of the source video—camera stability, coverage completeness, lighting consistency—has a far greater impact on reconstruction quality than the choice of training algorithm or parameter settings.
5. **Containerised, AI-orchestrated pipelines reduce the expertise barrier** for cultural heritage practitioners, encoding specialist knowledge into automated workflows that can be operated by non-specialists.
6. **The splat-as-artefact paradigm** offers a fundamentally different preservation philosophy: rather than converting captures into traditional 3D meshes (with attendant quality loss), the volumetric representation itself becomes the preservation target, retaining maximum scene information for future access and re-processing.

8.2 Value Proposition for the University of Salford

The findings of this pilot project position the University of Salford to lead in an emerging area of cultural heritage practice. The value proposition operates at multiple levels:

Institutional capability. The `gaussian-toolkit` provides the University with an operational capability for volumetric capture and preservation of its own cultural heritage activities—exhibitions, performances, installations, and events across its facilities.

Knowledge exchange. The pipeline and methodology developed in this project constitute transferable knowledge assets that can be shared with partner institutions, cultural organisations, and creative industry practitioners through the AI KE CoP framework.

Research positioning. The intersection of AI, 3D reconstruction, and cultural heritage is a rapidly growing research area with strong funding prospects. This pilot provides the preliminary data and demonstrated capability required for competitive research proposals.

Student and staff development. The technologies and workflows developed here provide

authentic, industry-relevant learning opportunities for students and staff in creative technologies, digital media, computer science, and cultural studies.
External engagement. Browser-based access to preserved cultural heritage content creates opportunities for public engagement, educational outreach, and international collaboration that extend the University's impact beyond its physical campus.

8.3 Next Steps and Timeline

We propose the following concrete next steps, organised into three phases:

8.3.1 Phase 1: Foundation (Q2–Q3 2026)

- Deploy the `gaussian-toolkit` on University of Salford infrastructure (local GPU server or managed cloud instance).
- Conduct capture methodology training for 5–10 cultural heritage practitioners within the University's partner network.
- Perform 3–5 additional test captures across diverse cultural heritage scenarios (different venue types, lighting conditions, artwork categories) to validate the pipeline's generalisability.
- Integrate the splat viewer with the University's digital collection management system as a proof-of-concept.
- Prepare a capture methodology quick-reference guide for distribution.

8.3.2 Phase 2: Institutional Rollout (Q4 2026 – Q2 2027)

- Establish a cultural heritage capture service, available to University departments and partner institutions, with documented workflows and quality assurance processes.
- Evaluate and integrate emerging mesh extraction methods (CoMe) and object segmentation (SAM3) as they become available.
- Develop a public-facing web portal for browsing and exploring preserved cultural heritage content.
- Initiate consortium development for EU Horizon Europe funding.

8.3.3 Phase 3: Scaling and Sustainability (Q3 2027 onwards)

- Submit Horizon Europe proposal with consortium partners.
- Deploy cloud-based processing infrastructure for multi-venue, multi-institution throughput.
- Establish data governance framework for long-term preservation of Gaussian splat archives.
- Develop VR/immersive viewing experiences using Unreal Engine for select high-profile collections.
- Publish research findings and contribute to emerging standards for volumetric cultural heritage data.

The pilot project documented in this report demonstrates that the technological foundations are sound, the workflow is practical, and the results are compelling. The path from pilot to institutional capability to international collaboration is clear. The University of Salford has an opportunity to establish itself at the forefront of AI-driven cultural heritage preservation—a field whose importance will only grow as the cultural sector's investment in immersive and experiential content continues to accelerate.

References

Blender Foundation (2024). *Blender — Free and Open 3D Creation Software*. URL: https://www.blender.org/.

Champion, E. (2011). 'Playing with the Past'. In: *Human–Computer Interaction Series*. DOI: 10.1007/978-1-84996-501-9.

Curless, B. and M. Levoy (1996). 'A Volumetric Method for Building Complex Models from Range Images'. In: *Proceedings of the 23rd Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pp. 303–312. DOI: 10.1145/237170.237269.

Fang, G., B. Wang, P. Chen, Z. Wang and Y. Lu (2024). 'Mini-Splatting: Representing Scenes with a Constrained Number of Gaussians'. In: *European Conference on Computer Vision (ECCV)*.

Federal Agencies Digital Guidelines Initiative (2016). *Technical Guidelines for Digitizing Cultural Heritage Materials*. URL: http://www.digitizationguidelines.gov/guidelines/.

Grand View Research (2024). *Immersive Experience Market Size, Share & Trends Analysis Report*. Grand View Research. URL: https://www.grandviewresearch.com/industry-analysis/immersive-experience-market.

Guédon, A. and V. Lepetit (2024). 'SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering'. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

— (2025). 'MILo: Mesh Extraction from Gaussian Splatting with Isosurface Lifting'. In: *arXiv preprint arXiv:2501.XXXXX*.

Kerbl, B., G. Kopanas, T. Leimkühler and G. Drettakis (2023). '3D Gaussian Splatting for Real-Time Radiance Field Rendering'. In: *ACM Transactions on Graphics (Proceedings of SIGGRAPH)*. Vol. 42. 4. DOI: 10.1145/3592433.

Kheradmand, S., D. Rebain, G. Sharma, W. Sun, J. Tseng, H. Isack, A. Kar, A. Tagliasacchi and K. M. Yi (2024). '3D Gaussian Splatting as Markov Chain Monte Carlo'. In: *Advances in Neural Information Processing Systems (NeurIPS)*.

Kirillov, A., E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar and R. Girshick (2023). 'Segment Anything'. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*.

Lorensen, W. E. and H. E. Cline (1987). 'Marching Cubes: A High Resolution 3D Surface Construction Algorithm'. In: *ACM SIGGRAPH Computer Graphics* 21.4, pp. 163–169. DOI: 10.1145/37402.37422.

Lowe, D. G. (2004). 'Distinctive Image Features from Scale-Invariant Keypoints'. In: *International Journal of Computer Vision* 60.2, pp. 91–110. DOI: 10.1023/B:VISI.0000029664.99615.94.

Mildenhall, B., P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi and R. Ng (2020). 'NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis'. In: *European Conference on Computer Vision (ECCV)*, pp. 405–421.

Mortara, M., C. E. Catalano, F. Bellotti, G. Fiucci, M. Houry-Panchetti and P. Petridis (2014). 'Learning Cultural Heritage by Serious Games'. In: *Journal of Cultural Heritage*. Vol. 15. 3, pp. 318–325. DOI: 10.1016/j.culher.2013.04.004.

Nocerino, E., F. Menna, F. Remondino, I. Toschi and P. Rodríguez-Gonzálvez (2017). 'Investigation of Indoor and Outdoor Performance of Two Portable Mobile Mapping Systems'. In: *Proceedings of SPIE* 10332. DOI: 10.1117/12.2270761.

Pine, B. J. and J. H. Gilmore (1998). *The Experience Economy: Work Is Theatre & Every Business a Stage*. Harvard Business School Press.

Pixar Animation Studios (2019). *Universal Scene Description*. URL: https://openusd.org/.

Radl, L., M. Steiner, M. Parger, B. Kerbl and M. Wimmer (2025). 'Confidence-Based Mesh Extraction from 3D Gaussian Splatting'. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Remondino, F. (2011). 'Heritage Recording and 3D Modeling with Photogrammetry and 3D Scanning'. In: *Remote Sensing* 3.6, pp. 1104–1138. DOI: 10.3390/rs3061104.

Schönberger, J. L. and J.-M. Frahm (2016). 'Structure-from-Motion Revisited'. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4104–4113. DOI: 10.1109/CVPR.2016.445.

Schönberger, J. L., E. Zheng, J.-M. Frahm and M. Pollefeys (2016). 'Pixelwise View Selection for Unstructured Multi-View Stereo'. In: *European Conference on Computer Vision (ECCV)*, pp. 501–518.

UNESCO (2003). *Convention for the Safeguarding of the Intangible Cultural Heritage*. Adopted 17 October 2003. URL: https://ich.unesco.org/en/convention.

Wang, Z., A. C. Bovik, H. R. Sheikh and E. P. Simoncelli (2004). 'Image Quality Assessment: From Error Visibility to Structural Similarity'. In: *IEEE Transactions on Image Processing* 13.4, pp. 600–612. DOI: 10.1109/TIP.2003.819861.

Zhang, R., P. Isola, A. A. Efros, E. Shechtman and O. Wang (2018). 'The Unreasonable Effectiveness of Deep Features as a Perceptual Metric'. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 586–595. DOI: 10.1109/CVPR.2018.00068.

Zhao, Z. et al. (2025). 'Hunyuan3D 2.0: Scaling Diffusion Models for High Resolution Textured 3D Assets Generation'. In: *arXiv preprint arXiv:2501.12202*.

Appendix A

Pipeline Configuration Reference

This appendix provides the full configuration parameters and Docker Compose specification for the `gaussian-toolkit` pipeline as deployed during this project.

A.1 Docker Compose Configuration

```yaml
version: '3.8'
services:
  gaussian-toolkit:
    image: gaussian-toolkit:latest
    build:
      context: .
      dockerfile: Dockerfile
    runtime: nvidia
    environment:
      - NVIDIA_VISIBLE_DEVICES=all
    ports:
      - "8080:8080"    # Web interface
      - "8081:8081"    # Splat viewer
    volumes:
      - ./data:/workspace/data
      - shared-vol:/workspace/shared

  milo-service:
    image: milo-sidecar:latest
    build:
      context: ./milo
      dockerfile: Dockerfile
    runtime: nvidia
    ports:
      - "8082:8082"    # MILo REST API
    volumes:
      - shared-vol:/workspace/shared

volumes:
  shared-vol:
```

Listing A.1: Docker Compose service definition (abbreviated).

A.2 COLMAP Parameters

A.3 3DGS Training Parameters

Table A.1: COLMAP configuration parameters used in the pilot.

Parameter	**Value**	**Notes**
Feature extractor	SIFT (GPU)	Default COLMAP SIFT
Max features per image	8192	Increased from default 4096
Matching strategy	Sequential	Window size = 10
Bundle adjustment solver	Ceres (iterative)	Default
BA refinement iterations	10	Default
Min track length	3	Minimum views per 3D point

Table A.2: 3DGS training parameters for MRNF and MCMC variants.

Parameter	**MRNF**	**MCMC**	**Notes**
Total iterations	30,000	30,000	
Learning rate (position)	0.00016	0.00016	Exponential decay
Learning rate (SH)	0.0025	0.0025	
Learning rate (opacity)	0.05	0.05	
Densify from iteration	500	500	
Densify until iteration	15,000	15,000	
Densify interval	100	100	
Opacity reset interval	3,000	N/A	MCMC uses different strategy
SH degree (max)	3	3	

A.4 Capture Methodology Quick-Reference

The following one-page reference summarises the capture methodology guidelines from Section 6.5 in a format suitable for distribution to cultural heritage practitioners.

Capture Methodology Quick Reference

Equipment

- Smartphone (iPhone 12+ or equivalent) or mirrorless camera
- Gimbal or handheld stabiliser (strongly recommended)
- Calibration target (printed checkerboard, A3 minimum)

Settings

- Resolution: 4K (3840 × 2160)
- Frame rate: 30 FPS
- Lock exposure and white balance before starting
- Disable HDR and any computational photography modes

Movement

- Walk at half normal pace or slower
- Keep the camera at chest height, roughly horizontal
- Move *smoothly*—no sudden pans, tilts, or direction changes
- Overlap: ensure every surface is visible from 3+ viewpoints

Coverage Pattern

1. Walk the perimeter of the space, facing inward
2. Walk a central path through the space
3. Capture individual objects of interest from 360° where possible
4. Record close-ups of fine details separately

Before You Leave

- Review footage for blur, gaps, or exposure problems
- Re-capture any problematic sections immediately
- Photograph the calibration target from 1–2 m distance
- Record ambient lighting conditions (notes or photo of light sources)

Appendix B

Quality Metrics Reference

This appendix defines the quality metrics used in Chapter 5 to evaluate reconstruction and mesh extraction quality.

B.1 Image Quality Metrics

B.1.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the ratio between the maximum possible signal power and the power of corrupting noise, expressed in decibels (dB). For two images I (reference) and $\hat{I}$ (reconstruction), each of size $M \times N$ pixels:

$$\mathrm{PSNR} = 10 \cdot \log_{10} \left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}} \right) \tag{B.1}$$

where MAX is the maximum pixel value (255 for 8-bit images) and

$$\mathrm{MSE} = \frac{1}{MN} \sum_{i=1}^{M} \sum_{j=1}^{N} \left[I(i,j) - \hat{I}(i,j) \right]^2 \tag{B.2}$$

Higher PSNR values indicate greater fidelity. For novel view synthesis, PSNR values above 25 dB are generally considered good quality, and above 30 dB excellent.

B.1.2 Structural Similarity Index (SSIM)

SSIM (Wang et al. 2004) measures the perceived similarity between two images by comparing luminance, contrast, and structural information:

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{B.3}$$

where μ_x, μ_y are pixel means, σ_x, σ_y are standard deviations, σ_{xy} is the covariance, and C_1, C_2 are stabilisation constants. SSIM ranges from -1 to 1, with 1 indicating perfect similarity.

B.1.3 Learned Perceptual Image Patch Similarity (LPIPS)

LPIPS (Zhang et al. 2018) uses deep neural network features (typically from a VGG or AlexNet backbone) to compute perceptual distance between images. Unlike PSNR and SSIM, LPIPS is trained on human perceptual judgements and correlates more closely with perceived visual quality. Lower LPIPS values indicate higher perceptual similarity.

B.2 Geometric Quality Metrics

B.2.1 Chamfer Distance

The Chamfer distance measures the average nearest-neighbour distance between two point sets S_1 and S_2:

$$d_{\mathrm{CD}}(S_1, S_2) = \frac{1}{|S_1|} \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2 + \frac{1}{|S_2|} \sum_{y \in S_2} \min_{x \in S_1} \|y - x\|_2 \tag{B.4}$$

This metric quantifies how closely a reconstructed mesh approximates the reference geometry. It is symmetric—measuring both the distance from reconstruction to reference and vice versa—ensuring that both missing geometry and spurious geometry are penalised.

B.2.2 Mean Reprojection Error

The mean reprojection error measures the average pixel distance between observed 2D feature locations and the projections of their corresponding triangulated 3D points through the estimated camera model. It is the primary quality metric for SfM reconstruction accuracy:

$$\bar{e} = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{x}_i - \pi(K, R, t, \mathbf{X}_i)\|_2 \tag{B.5}$$

where $\mathbf{x}_i$ is the observed 2D location, $\mathbf{X}_i$ is the triangulated 3D point, π is the projection function, and K, R, t are the camera intrinsics, rotation, and translation respectively.

Values below 1.0 pixel indicate a well-conditioned reconstruction; values above 2.0 pixels suggest systematic calibration errors or insufficient feature quality.